

DeepUL: Deep Unlearning via Model Sparsity

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Abstract

Deep neural networks (DNNs) have achieved remarkable success across various domains, but their reliance on large datasets often containing sensitive information raises significant privacy concerns. In response to stringent privacy regulations such as GDPR and CCPA, machine unlearning (MUL) has emerged as a critical technique to remove the influence of specific data samples from trained models. However, existing unlearning methods face two major challenges: privacy deficiencies in the remaining datasets and significant instability of the model after unlearning. To address these issues, we propose DeepUL, a deep unlearning algorithm that leverages model sparsity to enhance both privacy and stability. DeepUL follows a two-step process: first pruning and then unlearning. The pruning step not only sparsifies the model but also decouples the model parameters from the original training data, thereby improving privacy. The unlearning step employs gradient projection to eliminate dependencies on the data to be deleted while maintaining the model stability. Additionally, we introduce a hierarchical weight pruning strategy to achieve differential pruning across network layers, preventing layer collapse and ensuring robustness. Extensive experiments on benchmark datasets demonstrate that DeepUL outperforms existing methods in utility, unlearning efficacy, privacy guarantee, and stability, making it a practical and effective solution for machine unlearning.

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CCS Concepts

- **Computing methodologies** → **Machine learning approaches**;
- **Security and privacy** → *Social network security and privacy*.

Keywords

Machine Unlearning, DNN, Model Sparsity, Privacy Deficiency

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1 Introduction

Deep neural networks (DNNs) have widespread application across various domains of artificial intelligence (AI), and they are typically trained on vast datasets, which often include sensitive personal information [9]. But this practice has raised significant concerns regarding privacy, as individuals increasingly demand that their data be removed from both the training datasets and the already-trained models to mitigate potential misuse or exposure. Moreover, stringent privacy regulations such as the General Data Protection Regulation (GDPR) [19] and the California Consumer Privacy Act (CCPA) [18] have been enacted, enforcing that AI service providers comply with the "Right to be Forgotten" by removing users' training data upon request. To address this regulatory requirement, machine unlearning (MUL) techniques have emerged [1], designed to eliminate the influence of specific data samples from an already-trained model, thereby protecting user privacy.

From a technical standpoint, MUL approaches can be broadly categorized into two paradigms: exact unlearning and approximate unlearning. Exact unlearning demands that the model's parameter distribution after unlearning be identical to that of a model

retrained from the remaining data. One of the most straightforward approaches to achieve exact unlearning is SISA [1], which leverages data partitioning to facilitate efficient retraining. However, due to the computational complexity and resource-intensive nature of exact unlearning, researchers have increasingly focused on approximate unlearning, which seeks to approximate the effect of retraining through efficient operations while maintaining an acceptable margin of error. The approximate unlearning methods include mainly two categories: influence function-based unlearning methods and gradient of loss function-based unlearning methods. The first category of methods[8, 13] leverages the influence function to estimate the influence of the data to be deleted on the model parameters. The second category of method[27, 29] calculates the gradients contributed by the data to be deleted during the training process and updates the already-trained model by adding back these gradients to approximate the model that would be obtained by retraining from scratch.

In recent years, the integration of model sparsity has emerged as a promising breakthrough for approximate unlearning. Jia et al.[16] demonstrated that model sparsity can be effectively leveraged to facilitate unlearning by incorporating sparsity into the unlearning process. They found that model sparsity inherently benefits the privacy of machine unlearning without additional cost. Fan et al. [10] introduced SalUn, an unlearning method that achieves effective unlearning in both image classification and generation tasks. SalUn operates by identifying salient weights and fine-tuning them using data to be deleted, which is assigned with random labels and the remaining data. However, SalUn may inadvertently memorize information about the unlearned data by associating them with random labels. Furthermore, fine-tuning on the remaining dataset can strengthen the model's memory of the retained data, leading to severe privacy leakage. Wu et al.[28] proposed Scissorhands to erase the influence of data. Scissorhands identifies critical parameters related to data to be deleted through connection sensitivity analysis, destroys the key information, and reinitializes the top- k percent of these parameters. However, when a large amount of data is unlearned, Scissorhands may disrupt crucial model parameters, necessitating relearning operations to repair the model. This leads to model instability and significantly reduces the efficiency of the unlearning process. In summary, there are two critical challenges in existing unlearning methods:

- **Privacy deficiencies in remaining datasets.** Existing methods exhibit significant privacy deficiencies in the remaining datasets. While pruning techniques theoretically offer additional privacy protection for unlearning mechanisms, current implementations primarily focus on eliminating the influence of deleted data, neglecting their potential role in enhancing privacy. Moreover, existing pruning-based unlearning algorithms explicitly leverage the remaining data to repair the model. This methodology substantially intensifies the model's memorization intensity of the remaining data, causing exploitable vulnerabilities for membership inference attacks and consequently exacerbates the risk of privacy leakage.
- **Significant instability of the model after unlearning.** Existing methods are highly sensitive to the scale of data to

be deleted. Empirically, we observed that as the proportion of data to be deleted increases from 10% to 90%, the accuracy of the model after unlearning represents great instability. Additionally, hyperparameters, such as the sparsity threshold and retraining learning rate, are highly sensitive, where even minor changes can result in significant variations in unlearning effectiveness, further contributing to instability. This instability severely limits the robustness of unlearning methods across different data scales and hardware environments.

To address these challenges, we propose DeepUL, a deep unlearning algorithm via model sparsity. Our algorithm follows a two-step process: pruning first, then unlearning. The process begins with sparsification of the target model, which not only enhances the operational efficiency of the unlearning but also achieves effective decoupling between model parameters and original training data, thereby significantly improving the privacy of the unlearned model. For stability considerations, our algorithm exclusively manipulates gradients derived from the dataset to be deleted. Through gradient projection, we strategically eliminate dependencies on the dataset to be deleted while maintaining the stability of the unlearning process. This dual mechanism of parameter decoupling and gradient-space constraint optimization collectively ensures both provable completeness and robustness of our unlearning algorithm. Additionally, we introduce a hierarchical weight pruning strategy to achieve differential pruning control across different network layers, thereby avoiding the impact of hyperparameters on the model's stability. Our key contributions are summarized as follows:

- We propose the DeepUL that integrates two steps: pruning and unlearning. The pruning step not only facilitates model sparsification and unlearning but also enhances the privacy guarantee of the unlearned model. Subsequently, in the unlearning step, the pruned model can effectively unlearn the specified data. The synergistic effect of these two steps enables our algorithm to excel at the unlearning task while maintaining model accuracy.
- We employ hierarchical pruning to optimize the model structure, which applies different pruning rates to different layers of the deep models. This method effectively reduces computational overhead while maintaining the model stability, preventing the phenomenon of layer collapse, where excessive pruning in certain layers leads to a sharp decline in the overall model performance.
- We conduct extensive experiments on three benchmark datasets, comparing DeepUL with multiple baseline methods. The results demonstrate that DeepUL not only achieves effective unlearning but also outperforms existing unlearning methods in terms of utility, unlearning efficacy, privacy guarantee, and stability.

2 Related Work

In this section, we briefly review the related work of machine unlearning, including exact and approximate unlearning. We also emphasize the latest advancements in model sparsity in machine unlearning, which provide significant insights for the research presented in this paper.

2.1 Model Sparsity

Model sparsification and network pruning are closely related concepts in the field of DNNs, both aimed at reducing the complexity of neural networks while maintaining or even improving their performance. Model sparsification is often achieved through network pruning, which involves deriving a significantly smaller sub-network from a large reference neural network that can replicate the performance of the original model. The notable lottery ticket hypothesis (LTH) [11] demonstrated the existence of sparse sub-networks, referred to as ‘winning ticket’, which can match or even exceeds the test accuracy of the original dense model. In addition to generalization, the effects of pruning have been extensively studied in various domains, including robustness [6], fairness [22], interpretability [5], loss landscape [12], and privacy [3].

Particularly, the privacy benefits of pruning [25] suggest a connection between data influence and model sparsity. For instance, Wang et al. [26] proposed a pruning algorithm that prevents privacy leakage from membership inference attacks (MIAs) while maintaining competitive accuracy with the original DNNs. Furthermore, it has been proven that pruning a specific layer of a neural network is equivalent to adding a certain amount of differentially private noise to its hidden-layer activations.

2.2 Machine Unlearning

The concept of machine unlearning was first introduced by Bourtoule et al. [1] as a mechanism to eliminate the influence of specific data points on a pre-trained model. The primary objective of MUL is to negate the impact of a particular subset of training data points on a model while maintaining its utility for the remaining data. There are two categories of unlearning methods, exact unlearning and approximate unlearning.

Exact machine unlearning methods [1, 2] require retraining the model from scratch using the remaining dataset after the target data has been removed. The primary advantage of exact unlearning is that it ensures complete elimination of the influence of the removed data, as the retrained model is never exposed to the unlearned data during training. Despite its strong unlearning guarantees, exact unlearning typically can only be applicable for simple and well-structured models [20]. Additionally, it often requires the storage of the entire training dataset, which may pose challenges in scenarios such as Machine Learning as a Service (MLaaS), where storing training data is either impractical or prohibited due to data privacy regulations. These limitations restrict the applicability of exact unlearning for complex models and environments.

Approximate machine unlearning involves directly modifying the parameters of a already-trained model to obtain the unlearned model that approximates the model retrained from scratch. This is typically achieved by updating the parameters of already-trained models with a limited number of iterations using the information derived from the unlearned data. Some approximate unlearning methods [8, 13] leverage the influence function to estimate the influence of the data to be deleted on the model parameters. While they require only a single Newton update to remove the model contribution of the deleted data, they have significant limitations: (i) computing the inverse Hessian matrix of the loss function can be computationally expensive, particularly for large-scale deep models,

and (ii) they may not be feasible in scenarios where the training dataset is no longer stored or has been deleted.

Other approximate unlearning methods [27–29] calculate the gradients contributed by the data to be deleted during the training process and update the already-trained model by adding back these gradients to approximate the model that would be obtained by retraining from scratch.

Recent work by Liu et al. [16] introduced an approach to MUL through model sparsification via weight pruning, bridging the gap between exact unlearning and approximate unlearning. Specifically, they integrate a ℓ_1 norm-based sparse penalty into the unlearning loss function, which quantifies the memorization strength of the model parameters on the dataset to be deleted. By minimizing this objective loss function, the model’s memorization of the target dataset is effectively erased.

In the context of federated learning (FL), Wang et al. [24] explored the problem of selectively unlearning classes from trained convolutional neural network (CNN)-based classification models. They proposed a method for scrubbing the model cleanly of information about particular classes. Channels with high Term Frequency Inverse Document Frequency (TF-IDF) scores have more discrimination on the target classes and thus need to be pruned to unlearn. The channel pruning is followed by a fine-tuning process to recover the performance of the pruned model.

To identify a universal principle of machine unlearning applicable to both classification and generation tasks, Fan et al. [10] devised an MUL paradigm termed “weight saliency.” This concept redirects the machine’s unlearning attention towards influential weights rather than the entire model, thereby enhancing effectiveness and efficiency. SalUn is the first principled MUL approach adaptable enough to effectively erase the influence of data and classes to be deleted in both image classification and generation.

In summary, the integration of model sparsity and pruning techniques into machine unlearning has opened new avenues for achieving efficient and secure data removal. From privacy-preserving pruning to weight saliency, these approaches collectively advance the state-of-the-art in machine unlearning, offering robust solutions for various applications in DNNs.

3 Methodology

3.1 Overview

Let $\mathcal{M}$ denote an already-trained model with parameters $\theta \in \mathbb{R}^d$, where d represents the dimension of the parameter space. The model $\mathcal{M}$ is trained on a dataset $\mathcal{D} = \{\mathbf{x}_i, \mathbf{y}_i\}_{i=1}^N$, where $\mathbf{x}_i$ denotes the input features and $\mathbf{y}_i$ represents the corresponding labels. Suppose a subset of $\mathcal{D}$, denoted as $\mathcal{D}_f$, is requested for deletion. The remaining data is defined as $\mathcal{D}_r := \mathcal{D} \setminus \mathcal{D}_f$. Our objective is to develop an machine unlearning method that generates an unlearned model $\mathcal{M}_u$, effectively eliminating the influence of $\mathcal{D}_f$ from model $\mathcal{M}$, while maintaining the utility of $\mathcal{M}$ over $\mathcal{D}_r$. To achieve the objective, our DeepUL, as illustrated in Fig. 1, operates in two main phases: pruning and unlearning.

In the pruning phase, we firstly identify the key parameters of the model $\mathcal{M}$. Within each network layer, the weight parameters of $\mathcal{M}$ are ranked based on their magnitudes. A pruning strategy is then applied, where important weight parameters are retained according

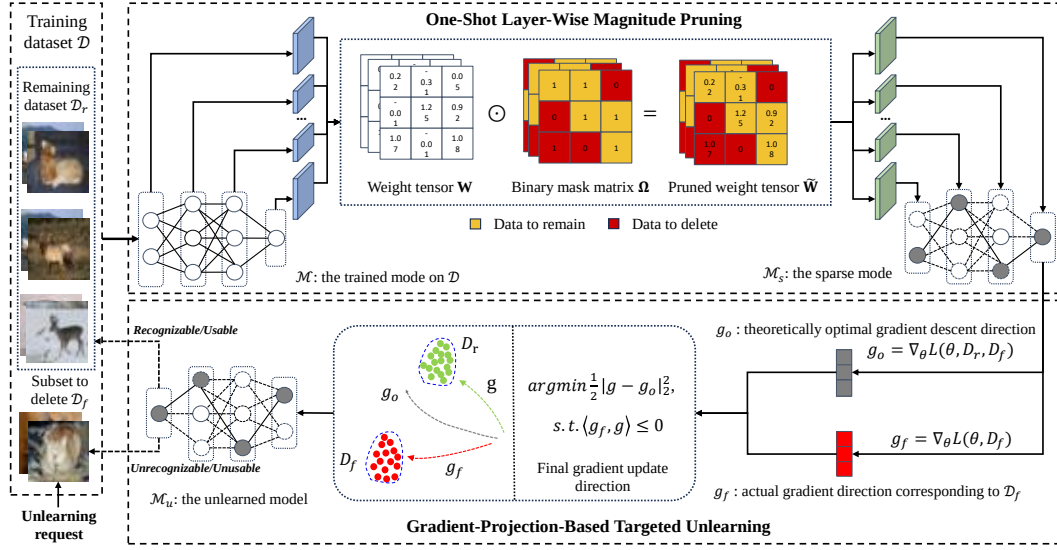


Figure 1: Overview of DeepUL. DeepUL operates in two phases: one-shot layer-wise magnitude pruning (OLMP) and gradient-projection-based targeted unlearning (GPTU). OLMP retains important weight parameters within each layer and prunes less significant ones, thereby extracting a trainable sparse submodel from the original dense model. GPTU uses a gradient projection-based method that manipulates the gradients derived from the dataset to be deleted and projects them in a way that minimizes their influence on the model parameters, obtaining an unlearned model.

to a predefined pruning rate, while less significant ones are pruned. This pruning process aims to extract a trainable sparse submodel $\mathcal{M}_s$ from the original dense model, ensuring that the utility of the pruned model is comparable to or even superior to that of the original model. By pruning that achieves model sparsification, we not only retain the accuracy and usability of the pruned model, but also ensure that its performance on both the training and test sets remains robust.

In the second phase of unlearning, we focus on effectively eliminating the influence of the dataset $\mathcal{D}_f$ while preserving the performance of $\mathcal{M}$ on the retained dataset $\mathcal{D}_r$. This dual objective is achieved through a gradient projection-based optimization approach, which carefully balances the trade-off between data unlearning and model accuracy. Specifically, we manipulate the gradients derived from the dataset $\mathcal{D}_f$ and project them in a way that minimizes their influence on the model parameters. This ensures that the model "forgets" the dataset $\mathcal{D}_f$.

As a result, we obtain an unlearned model $\mathcal{M}_u$, which effectively removes the influence of $\mathcal{D}_f$ without compromising the utility of $\mathcal{M}_u$.

3.2 One-Shot Layer-Wise Magnitude Pruning

To effectively identify the important model parameters, it is essential to establish a robust criterion. The magnitude-based evaluation criterion is widely adopted due to its simplicity and effectiveness: the absolute value of a weight is positively correlated with its contribution to the model, implying that parameters with smaller absolute values contribute relatively less to the model. We propose a layer-wise magnitude pruning strategy, referred to as One-Shot Layer-wise Magnitude Pruning (OLMP), which removes weights

with small absolute values in a single step. Therefore, OLMP ensures that critical layers are not overly pruned, thereby maintaining model performance. Furthermore, OLMP achieves a fine-grained balance between accuracy and efficiency during the model sparsification process by differentiating the sparsity levels across layers.

Given the weight tensor $\mathbf{W}^{(l)} \in \mathbb{R}^{m \times n}$ of the l -th layer in the already-trained model $\mathcal{M}$, we construct a binary mask matrix $\Omega^{(l)} \in \{0, 1\}^{m \times n}$. The pruned weight matrix is obtained through element-wise multiplication: $\tilde{\mathbf{W}}^{(l)} = \mathbf{W}^{(l)} \odot \Omega^{(l)}$. The objective function can be formalized as:

$$\min_{\Omega} \left\| \mathbf{W}^{(l)} - \tilde{\mathbf{W}}^{(l)} \right\|_F^2, \text{ s.t. } \frac{\|\Omega^{(l)}\|_0}{\|\Omega^{(l)}\|} \leq k_l \quad (1)$$

where $\|\cdot\|_F$ is the Frobenius norm, which quantifies the deviation between original and pruned weight matrices, and $\|\Omega^{(l)}\|_0$ and $\|\Omega^{(l)}\|$ represents the number of non-zero elements and the total number of elements in the mask, respectively, controlling the number of weights retained after pruning. The parameter k_l specifies the target sparsity level for layer l of the model $\mathcal{M}$.

To determine the optimal mask matrix, OLMP scans the weight distribution layer by layer, and sorts the parameters in descending order of their absolute values within each layer. According to the predefined pruning target k_l , a set of critical parameters S is conducted, which contains the parameters that need to be retained. A binary sparse mask matrix is constructed by assigning a value of 1 to the positions corresponding to the critical parameters (indicating retention) and setting all other positions to 0 (indicating pruning). By element-wise multiplication between the weight matrix and the mask matrix, non-critical parameters are precisely zeroed out while the critical ones are retained.

Through the aforementioned pruning process, OLMP effectively decouples the remaining data from the model, thereby enhancing model security by reducing dependency on redundant parameters and simultaneously enhancing its generalization capability. Moreover, this pruning process systematically reduces computational complexity by optimizing the model structure, establishing a streamlined framework for subsequent unlearning operations with significantly improved efficiency.

3.3 Gradient-Projection-Based Targeted Unlearning

After parameter pruning to derive a sparse submodel that maintains comparable performance to the original model, the key challenge lies in effectively erasing the memorization of the dataset $\mathcal{D}_f$ while preserving the informational integrity of the remaining dataset $\mathcal{D}_r$. This task is particularly challenging. The adversarial interplay between these two loss functions facilitates selective removal of data without compromising the model's core functionality. To achieve this targeted unlearning, we propose gradient-projection-based targeted unlearning (GPTU) that simultaneously maximizes the loss function for $\mathcal{D}_f$ and minimizes the loss function associated with $\mathcal{D}_r$ during parameter updates. Therefore, we formulate the unlearning as follows:

$$L(\theta, \mathcal{D}_r, \mathcal{D}_f) := \mathcal{L}(\theta; \mathcal{D}_r) - \lambda \mathcal{L}(\theta; \mathcal{D}_f). \quad (2)$$

where λ is a hyperparameter that controls the trade-off between retaining utility on $\mathcal{D}_r$ and removing $\mathcal{D}_f$.

To optimize Equation (2) with respect to θ , GPTU employ a gradient projection-based approach inspired by [28]. The theoretically optimal gradient descent direction, denoted as $g_o = \nabla_{\theta} L(\theta, \mathcal{D}_r, \mathcal{D}_f)$, is computed based on the loss function defined in Equation (2). To achieve this, we impose a directional constraint requiring the actual update direction g to remain non-congruent with g_f , where $g_f = \nabla_{\theta} L(\theta; \mathcal{D}_f)$ represents the gradient direction corresponding to dataset $\mathcal{D}_f$. This constraint is mathematically formalized as $\langle g_f, g \rangle \leq 0$, which geometrically enforces an angle of at least 90 degrees between the two gradient vectors in the parameter space. This constrained optimization problem can be efficiently solved using the Frank-Wolfe algorithm[7]. To simplify the computational processes and enhance efficiency, we adopt the dual formulation of the problem:

$$\begin{aligned} \sup_v \quad & \frac{1}{2} g_f^{\top} g_f v^2 - g_o^{\top} g_f v, \\ \text{s.t.} \quad & v \geq 0 \end{aligned} \quad (3)$$

where v is the dual variable that we aim to optimize, The dual formulation allows us to solve the problem more efficiently by leveraging the properties of convex optimization.

Then, we can obtain the optimal dual variable v^* , and the optimal descent direction g is computed as $g = g_o - v^* g_f$. Our GPTU ensures that the model update not only the performance-optimal direction g_o but also effectively mitigates the model's memorization of $\mathcal{D}_f$ by subtracting a component of g_f . Therefore, GPTU provides a mathematically rigorous and computationally efficient method for achieving the objectives of data unlearning and utility remaining of the unlearned model.

4 Experiments And Results

In this section, we present a comprehensive evaluation of the proposed DeepUL on tasks of random sample removal and class-wise removal. The experiments are conducted on three benchmark datasets. We compare the performance of DeepUL against two kinds of baselines: model retraining and eight state-of-the-art unlearning methods. Our results demonstrate the effectiveness and versatility of DeepUL in achieving efficient and accurate unlearning while maintaining model utility.

4.1 Experiment Setup

4.1.1 Datasets. The evaluation covers three benchmark datasets with distinct characteristics:

- **CIFAR-10** [15]: CIFAR-10 contains 60,000 32×32 color images (50,000 for training and 10,000 for testing) covering 10 object categories, providing a balanced class distribution for medium-scale classification.
- **CIFAR-100** [15]: CIFAR-100 extends the CIFAR-10 paradigm to fine-grained classification, comprising 100 distinct object categories organized hierarchically into 20 superclasses. Each category contains 600 images (500 for training and 100 for testing) with 32×32 pixel resolution.
- **SVHN** [17]: SVHN contains 73,257 training and 26,032 test images of 10-digit classes (0-9), featuring real-world house number photos with varying orientations and background clutter.

4.1.2 Baseline Methods. We compare DeepUL against nine established machine unlearning approaches, which are widely recognized in the literature. These baselines provide a comprehensive evaluation framework for assessing the effectiveness of DeepUL.

- **Retraining:** The Model is retrained from scratch using only the remaining data $\mathcal{D}_r$. This approach serves as an upper bound of the performance of different machine unlearning methods, as it represents the ideal scenario where the model has no influence from the data to be deleted.
- **Fine-tuning (FT)**[27]: The model is fine-tuned on $\mathcal{D}_r$, leveraging the phenomenon of catastrophic forgetting in deep neural networks to facilitate the unlearning process.
- **Gradient ascent (GA)**[23] GA is an optimization technique that aims to maximize a function by moving in the direction of its positive gradient. Unlike Gradient Descent (GD), which minimizes the loss, GA reverses the training process on the data to be deleted $\mathcal{D}_f$ by adding the gradient of the loss with respect to $\mathcal{D}_f$ back to the model parameters. This effectively increases the loss for $\mathcal{D}_f$, allowing the model to "forget" the data while maintaining its performance on the remaining data $\mathcal{D}_r$.
- **Influence unlearning (IU)** [14]: IU leverages the influence functions to approximate the changes in model parameters when transitioning the original model to the scrubbed.
- **Boundary shrink (BS)** [4] and **Boundary expanding (BE)**: They shrink or expanding the decision boundary of the original model to mimic that of the retrained model, effectively simulating the effects of retraining without the need for a full model retraining.

Table 1: The utility of unlearned models through different unlearning algorithms for sample-wise removal.

Methods	CIFAR-10				CIFAR-100				SVHN			
	TA($\uparrow$)	RA($\uparrow$)	UA($\downarrow$)	MIA-Efficacy ($\uparrow$)	TA($\uparrow$)	RA($\uparrow$)	UA($\downarrow$)	MIA-Efficacy ($\uparrow$)	TA($\uparrow$)	RA($\uparrow$)	UA($\downarrow$)	MIA-Efficacy ($\uparrow$)
Retrain	94.26	99.99	94.81	13.05	74.69	99.98	75.13	50.22	91.17	97.73	91.81	15.74
FT	93.83	99.84	99.15	3.01	75.28	99.95	97.98	9.64	95.28	99.99	99.60	3.85
GA	94.57	99.45	99.64	1.3	75.59	98.24	98.00	5.00	92.87	98.52	98.41	9.96
IU	91.91	98.01	98.08	4.01	72.13	96.14	95.67	9.44	86.92	93.36	92.47	17.00
BE	93.97	99.41	99.41	16.16	74.16	98.12	97.94	7.60	92.62	98.44	98.48	6.77
BS	94.24	99.56	99.60	4.46	73.20	97.93	97.65	8.24	92.48	98.36	98.29	24.47
ℓ_1 -sparse	91.05	97.47	94.68	11.87	69.99	96.21	87.89	21.33	93.59	99.49	97.79	7.57
SalUn	93.73	99.89	95.95	17.62	69.92	99.40	83.95	74.13	93.59	97.41	96.29	24.47
Scissorhands	92.92	98.97	97.34	9.56	73.60	98.14	69.37	43.79	93.05	99.51	91.62	17.62
DeepUL	93.73	99.36	93.26	10.9	72.85	99.66	80.02	32.55	93.90	99.64	90.54	16.19

Table 2: The utility of unlearned models through different unlearning algorithms for class-wise removal.

Methods	CIFAR-10				CIFAR-100				SVHN			
	TA($\uparrow$)	RA($\uparrow$)	UA($\downarrow$)	MIA-Efficacy ($\uparrow$)	TA($\uparrow$)	RA($\uparrow$)	UA($\downarrow$)	MIA-Efficacy ($\uparrow$)	TA($\uparrow$)	RA($\uparrow$)	UA($\downarrow$)	MIA-Efficacy ($\uparrow$)
Retrain	94.83	99.99	0	100	74.77	99.97	0	100	95.72	98.93	0	100
FT	94.27	99.94	92.35	44.86	74.75	99.82	46.88	92.66	95.24	99.99	82.68	93.71
GA	86.80	92.64	5.4	96.28	66.91	92.02	30.22	89.55	90.21	94.97	3.64	99.37
IU	91.42	97.98	12.18	95.96	71.24	96.23	15.88	98.44	91.73	97.16	77.09	99.89
BE	91.45	98.21	19.37	92.28	60.61	83.03	8.66	98.22	91.32	96.87	11.78	97.85
BS	91.53	98.23	19.00	92.06	60.61	83.03	8.66	98.22	91.21	96.75	11.76	97.56
ℓ_1 -sparse	91.08	97.06	0	100	67.45	92.38	0	100	92.37	99.34	0	100
SalUn	85.43	90.30	0	100	57.29	72.08	0	100	95.68	100	0	100
Scissorhands	94.55	99.97	0	100	67.14	78.51	0	100	95.05	99.82	0	100
DeepUL	91.75	97.72	0	100	73.48	98.03	0	100	94.47	99.89	0	100

- **ℓ_1 -sparse**[16]: The model is fine-tuned on $\mathcal{D}_r$ with ℓ_1 -norm sparse regularization. The ℓ_1 -norm encourages sparsity in the model parameters, which helps in reducing the influence of the deleted data.
- **Saliency unlearning (SalUn)** [10]: SalUn adopts weight saliency and random labeling to facilitate the unlearning process. By identifying and modifying the most influential parameters, SalUn effectively reduces the model's dependence on the data to be deleted.
- **Scissorhands** [28]: Scissorhands identifies the most critical parameters associated with the data to be deleted by measuring their connection sensitivity. The top- k percent of these highly influential parameters are then reinitialized.

4.1.3 Metrics. To comprehensively evaluate the performance of our DeepUL, we employ five key metrics that assess the unlearning efficacy and model accuracy. These metrics provide a holistic view of the model's ability to remove specific data while maintaining its utility and privacy guarantees.

- **Testing Accuracy (TA):** TA assesses the generalization capability of the unlearned model on a held-out testing dataset. TA is computed over the entire test dataset, except in cases

of class-wise removal, where samples belonging to the removed class are excluded from the testing scope. An effective unlearned model should maintain comparable accuracy to that of a retrained model, ensuring that the unlearning process does not significantly degrade the model's accuracy on unseen data. This metric is critical for evaluating the utility preservation and the generalization ability of the unlearned model.

- **Remaining Accuracy (RA):** RA measures the accuracy of the unlearned model on the remaining dataset $\mathcal{D}_r$. This metric reflects the fidelity of the unlearning process, ensuring that the unlearned model retains useful information from the original training data. A high RA indicates that the unlearning process has not adversely affected the model's accuracy. This is essential for maintaining the model's performance on the data that was not targeted for removal.
- **Unlearning Accuracy (UA):** UA measures the effectiveness of unlearning process and is defined by $UA(\theta_u) = \text{Acc}\mathcal{D}_f(\theta_u)$, where $\text{Acc}\mathcal{D}_f(\theta_u)$ denotes the accuracy of the unlearned model on the removed dataset $\mathcal{D}_f$. A lower UA indicates that the model has successfully "forgotten" the

specified data, which is particularly crucial for approximate machine unlearning methods. This metric ensures that the unlearning process effectively reduces the model's reliance on the removed data.

- **Membership Inference Attack Efficacy (MIA-Efficacy):** This metric evaluates the success rate of the unlearning process by applying a confidence-based membership inference attack (MIA) predictor [21] to the unlearned model on the removed dataset $\mathcal{D}_f$. The MIA success rate indicates the proportion of samples in $\mathcal{D}_f$ that are correctly identified as non-training samples by the unlearned model. A higher MIA-Efficacy implies that the unlearned model retains less information about $\mathcal{D}_f$, thereby demonstrating the success of the unlearning process.
- **Membership Inference Attack Privacy (MIA-Privacy):** MIA-Privacy evaluates the privacy guarantees of an unlearned model with respect to the retained dataset $\mathcal{D}_r$. Specifically, it measures the extent to which an adversary can infer whether a particular data point was part of the original training dataset $\mathcal{D}$ or not. A lower MIA-Privacy score indicates that the unlearned model has less information leakage, thereby providing stronger privacy protection for the retained data. This metric is crucial for ensuring that the unlearning process does not compromise the privacy of the remaining data.

4.2 Unlearning Efficacy

We evaluate the efficacy of unlearned models for both sample-wise removal and class-wise removal. The results focus on the TA, RA, UA and MIA-Efficacy. The comparative results are presented in Table 1 and Table 2, respectively.

For sample-wise removal, on CIFAR-10 DeepUL achieves a TA of 93.73% and an RA of 99.36%, which is only 0.49% and 0.63% lower than Retrain, respectively. These small gaps indicate that DeepUL can achieve near-retrain performance, making it a highly efficient alternative. Other algorithms such as FT and Scissorhands also show strong performance, particularly in terms of TA and RA. However, DeepUL exhibits the lowest UA among the compared methods. DeepUL achieves a UA of 93.26% that is significantly lower than other algorithms such as FT (99.15%) and GA (99.64%) and a MIA-Efficacy of 10.9% that is close to that of Retrain (13.05%). This indicates that DeepUL is more effective at removing the influence of the deleted samples. On SVHN, DeepUL achieves the lowest UA of 90.54% and a MIA-Efficacy of 16.19%, which is close to that of Retrain (15.74%).

For class-wise removal, the results are more nuanced. As shown in Table 2, FT, Scissorhands, and DeepUL all exhibit strong utility in terms of TA and RA. However, FT's performance comes at the cost of highest UA, indicating that it retains too much information from the forgotten class. In contrast, DeepUL achieves perfect unlearning, with a UA of 0% and an MIA-Efficacy of 100%. This demonstrates DeepUL's ability to effectively balance utility (maintaining model accuracy) and unlearning effectiveness (removing the influence of the deleted class).

4.3 Privacy Guarantee of Unlearned Model

We evaluate the privacy guarantee of the unlearned model by the MIA-privacy metric, the results are shown in Table 3 and Table 4.

In the context of sample-wise removal, the ℓ_1 -sparse and DeepUL algorithms consistently outperform other methods across all three datasets. For instance, on CIFAR-10, the ℓ_1 -sparse algorithm achieves the lowest MIA-Privacy score of 53.92, closely followed by DeepUL with a score of 54.54. Both algorithms significantly outperform the baseline Retrain method, which has a score of 59.13. This suggests that model sparsity plays a crucial role in enhancing privacy protection by decoupling the remaining data from the unlearned model, thereby reducing the risk of MIA attacks. Similarly, on CIFAR-100, ℓ_1 -sparse algorithm again leads with a score of 65.17, followed by Scissorhands (67.70) and DeepUL (68.82). On SVHN, DeepUL and IU algorithms show strong privacy protection capabilities, with DeepUL achieving a score of 54.05. This suggests that DeepUL is especially effective in simpler datasets, where it maintains a balance between model utility and privacy.

In contrast, the results for class-wise removal are less pronounced, but certain algorithms still demonstrate strong privacy protection capabilities. On CIFAR-10 and CIFAR-100, DeepUL performs slightly worse, but still outperforms the baseline Retrain method. On SVHN, BE and BS algorithms again perform well with a score of 51.40, followed by GA (54.07) and DeepUL (54.08). This suggests that while DeepUL is effective, other algorithms like BE and BS may be more suitable for class-wise removal in simpler datasets. The less pronounced results in class-wise removal could be due to the inherent complexity of removing entire classes from the model, which may not be as effectively addressed by sparsity-based methods.

The comparative analysis reveals that although DeepUL fails to achieve the best MIA-Privacy scores, DeepUL offers a more balanced approach by maintaining model utility, as indicated by TA and RA metrics. This makes DeepUL a more practical choice where both privacy and model performance are critical. Additionally, algorithms like BE, BS, and GA show strong performance in class-wise removal, particularly in simpler datasets like SVHN.

Table 3: The MIA-Privacy of different unlearning algorithm for removing 10% of samples from datasets

Methods	CIFAR-10	CIFAR-100	SVHN
	MIA-Privacy($\downarrow$)	MIA-Privacy($\downarrow$)	MIA-Privacy($\downarrow$)
Retrain	59.13	73.45	54.32
FT	55.13	71.11	56.64
GA	55.80	70.02	54.72
IU	55.75	69.05	51.93
BE	55.57	69.92	54.77
BS	55.70	69.95	54.76
ℓ_1 -sparse	53.92	65.17	54.90
SalUn	56.83	71.43	57.53
Scissorhands	56.04	67.70	55.42
DeepUL	54.54	68.82	54.05

4.4 Unlearning Stability

Finally, we evaluate the stability of DeepUL against other state-of-the-art unlearning algorithms. The experiments are conducted on

Table 4: The MIA-Privacy of different unlearning algorithm for removing 10% of classes from datasets

Methods	CIFAR-10	CIFAR-100	SVHN
	MIA-Privacy($\downarrow$)	MIA-Privacy($\downarrow$)	MIA-Privacy($\downarrow$)
Retrain	55.47	70.76	54.35
FT	55.07	69.68	56.36
GA	52.80	63.90	54.07
IU	55.06	67.61	54.20
BE	53.85	61.10	51.40
BS	53.84	61.10	51.40
ℓ_1 -sparse	54.16	63.39	54.57
SalUn	52.91	70.02	56.92
Scissorhands	51.38	66.54	55.37
DeepUL	54.76	68.68	54.08

the CIFAR-10 dataset, and the results are analyzed in terms of test accuracy (TA) and robustness to changes in data distribution.

In the sample-wise removal experiments, DeepUL demonstrates remarkable stability across a wide range of data removal ratios, from 10% to 80%. As shown in Table 5, DeepUL maintains a high TA, with only a 3.29% drop when the removal ratio increases from 10% to 80%. Specifically, the TA decreases from 93.36% to 90.62%, which is a minimal degradation compared to other unlearning algorithms. The TA of the Scissorhands algorithm suffers plummeted to 12.88% when 80% of the samples are removed, indicating a significant flaw in its parameter pruning strategy. This stark contrast highlights the robustness of DeepUL’s pruning strategy, which selectively retains critical model parameters while pruning less important ones.

The class-wise removal experiments further validate the stability of DeepUL. As depicted in Table 6, DeepUL’s TA remains consistently high, even when up to 30% of the classes are removed. At 30% class removal ratio, DeepUL achieves a TA of 73.93%, which is significantly higher than the 71.65% achieved by ℓ_1 -sparse and the 69.87% achieved by SalUn. Notably, DeepUL’s performance is close to that of the retrained model, which achieves 73.25% under the same conditions. This suggests that DeepUL effectively maintains the model’s generalization ability even when a substantial portion of the data is removed. Scissorhands exhibits a sharp decline in test accuracy, dropping to 50.40% at a 30% class removal ratio. The ℓ_1 -sparse algorithm, while more stable than Scissorhands, still falls short of DeepUL’s performance, particularly at higher removal ratios. These findings underscore the importance of DeepUL’s strategy of protecting core model features during the unlearning process, which ensures both stability and utility.

Through comprehensive evaluation of multiple advanced unlearning algorithms, DeepUL consistently demonstrates superior performance across four critical dimensions: utility, unlearning efficacy, privacy guarantee, and stability. While other algorithms may excel in isolated metrics, DeepUL’s systematic and multi-faceted approach ensures robust performance under varying data removal scenarios. These results underscore DeepUL’s balance to model sparsity and feature retention, positioning it as a leading solution for stable and effective machine unlearning.

Table 5: TA of different unlearning methods in removing varying ratios of data samples from the CIFAR-10 dataset.

Methods	Data sample removal ratios			
	10%	30%	50%	80%
Retrain	94.26	93.64	91.72	85.69
ℓ_1 -sparse	90.64	90.17	90.78	91.27
SalUn	93.73	91.89	89.93	88.77
Scissorhands	92.92	93.02	85.04	12.88
DeepUL	93.36	93.91	91.30	90.62

Table 6: TA of different unlearning methods in removing varying ratios of class samples from the CIFAR-10 dataset.

Methods	Class sample removal ratios		
	10%	20%	30%
Retrain	94.27	84.26	73.25
ℓ_1 -sparse	91.08	80.54	71.65
SalUn	85.43	78.65	69.87
Scissorhands	94.55	65.28	50.40
DeepUL	91.75	83.66	73.93

5 Conclusion

In this work, we propose DeepUL, an effective and practical machine unlearning algorithm that significantly improves unlearning performance in terms of model usability, unlearning efficacy, security, and stability. Our algorithm adopts a prune-first-then-unlearn strategy, where the model is first pruned to obtain a sparse sub-model, and then unlearning is achieved through gradient projection within the pruned model. This two-step approach not only enhances the privacy of the unlearned model by decoupling it from the original training data but also ensures stability by avoiding excessive parameter disruption. Furthermore, our hierarchical pruning strategy applies different pruning rates across network layers, preventing layer collapse. Extensive evaluations on CIFAR-10, CIFAR-100, and SVHN demonstrate that DeepUL outperforms existing methods in both scenarios of sample-wise removal and class-wise removal. Compared to other approaches, DeepUL provides superior performance in terms of utility, unlearning efficacy, privacy guarantee of the unlearned model, and unlearning stability. These results highlight the effectiveness of DeepUL as a robust and practical machine unlearning solution for deep neural networks, addressing the critical challenges of privacy and stability in modern AI systems.

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References

- [1] Lucas Bourtole, Varun Chandrasekaran, Christopher A Choquette-Choo, Hengrui Jia, Adelin Travers, Baiwu Zhang, David Lie, and Nicolas Papernot. 2021. Machine unlearning. In *Proceedings of the 2021 IEEE Symposium on Security and Privacy (SP'21)*. IEEE, 141–159.
- [2] Jonathan Brophy and Daniel Lowd. 2021. Machine unlearning for random forests. In *Proceedings of the 38th International Conference on Machine Learning (ICML'21)*. PMLR, 1092–1104.
- [3] Yifei Cai, Qiao Zhang, Rui Ning, Chunsheng Xin, and Hongyi Wu. 2022. Hunter: HE-Friendly Structured Pruning for Efficient Privacy-Preserving Deep Learning. In *Proceedings of the 2022 ACM on Asia Conference on Computer and Communications Security (ASIA CCS '22)*, 931–945.
- [4] Min Chen, Weizhuo Gao, Gaoyang Liu, Kai Peng, and Chen Wang. 2023. Boundary unlearning: Rapid forgetting of deep networks via shifting the decision boundary. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR'23)*, 7766–7775.
- [5] Tianlong Chen, Zhenyu Zhang, Jun Wu, Randy Huang, Sijia Liu, Shiyu Chang, and Zhangyang Wang. 2022. Can you win everything with a lottery ticket? *Transactions on Machine Learning Research* (2022).
- [6] Tianlong Chen, Zhenyu Zhang, Yihua Zhang, Shiyu Chang, Sijia Liu, and Zhangyang Wang. 2022. Quarantine: Sparsity can uncover the trojan attack trigger for free. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR'22)*, 598–609.
- [7] Farah Cherfaoui, Valentin Emiya, Liva Ralaivola, and Sandrine Anthoine. 2019. Recovery and Convergence Rate of the Frank–Wolfe Algorithm for the m-Exact-Sparse Problem. *IEEE Transactions on Information Theory* 65, 11 (2019), 7407–7414.
- [8] R Dennis Cook and Sanford Weisberg. 1980. Characterizations of an empirical influence function for detecting influential cases in regression. *Technometrics* 22, 4 (1980), 495–508.
- [9] Emiliano De Cristofaro. 2021. A critical overview of privacy in machine learning. *IEEE Security & Privacy* 19, 4 (2021), 19–27.
- [10] Chongyu Fan, Jiancheng Liu, Yihua Zhang, Eric Wong, Dennis Wei, and Sijia Liu. 2024. SALUN: Empowering Machine unlearning via Gradient-Based Weight Saliency in Both Image Classification and Generation. In *Proceedings of the 20th International Conference on Learning Representations (ICLR'24)*.
- [11] Jonathan Frankle and Michael Carbin. 2019. The lottery ticket hypothesis: Finding sparse, trainable neural networks. In *Proceedings of the 7th International Conference on Learning Representations (ICLR'19)*.
- [12] Jonathan Frankle, Gintare Karolina Dziugaite, Daniel Roy, and Michael Carbin. 2020. Linear mode connectivity and the lottery ticket hypothesis. In *Proceedings of the 37th International Conference on Machine Learning (ICML'20)*. PMLR, 3259–3269.
- [13] Zachary Izzo, Mary Anne Smart, Kamalika Chaudhuri, and James Zou. 2021. Approximate data deletion from machine learning models. In *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics*. PMLR, 2008–2016.
- [14] Pang Wei Koh and Percy Liang. 2017. Understanding black-box predictions via influence functions. In *Proceedings of the 34th International Conference on Machine Learning (ICML'17)*. PMLR, 1885–1894.
- [15] Alex Krizhevsky, Geoffrey Hinton, et al. 2009. Learning multiple layers of features from tiny images. *Technical Report TR-2009* (2009).
- [16] Jiancheng Liu, Parikshit Ram, Yuguang Yao, Gaowen Liu, Yang Liu, PRANAY SHARMA, Sijia Liu, et al. 2024. Model Sparsity Can Simplify Machine Unlearning. In *Proceedings of the 37th International Conference on Neural Information Processing Systems (NIPS '23)*, Vol. 36, 51584 – 51605.
- [17] Yuval Netzer, Tao Wang, Adam Coates, Alessandro Bissacco, Baolin Wu, Andrew Y Ng, et al. 2011. Reading digits in natural images with unsupervised feature learning. In *NeurIPS Workshop on Deep Learning and Unsupervised Feature Learning*, Granada, 4.
- [18] Stuart L Pardau. 2018. The california consumer privacy act: Towards a european-style privacy regime in the united states. *J. Tech. L. & Pol'y* 23 (2018), 68.
- [19] Jeffrey Rosen. 2011. The right to be forgotten. *Stan. L. Rev. Online* 64 (2011), 88.
- [20] Sebastian Schelter. 2020. "Amnesia" - Machine Learning Models That Can Forget User Data Very Fast. In *Proceedings of the 10th Conference on Innovative Data Systems Research (CIDR'20)*. CIDR Proceedings.
- [21] Liwei Song, Reza Shokri, and Prateek Mittal. 2019. Privacy risks of securing machine learning models against adversarial examples. In *Proceedings of the 2019 ACM SIGSAC Conference on Computer and Communications Security (CCS'19)*, 241–257.
- [22] Samuil Stoychev and Hatice Gunes. 2022. The effect of model compression on fairness in facial expression recognition. In *Proceedings of the 26th International Conference on Pattern Recognition (ICPR'22)*. Springer, 121–138.
- [23] Anvith Thudi, Gabriel Deza, Varun Chandrasekaran, and Nicolas Papernot. 2022. Unrolling SGD: Understanding Factors Influencing Machine Unlearning. In *Proceedings of the 7th European Symposium on Security and Privacy (EuroSP'22)*. IEEE, 303–319.
- [24] Junxiao Wang, Song Guo, Xin Xie, and Heng Qi. 2022. Federated unlearning via class-discriminative pruning. In *Proceedings of the ACM Web Conference 2022 (WWW'22)*, 622–632.
- [25] Yijue Wang, Chenghong Wang, Zigeng Wang, Shanglin Zhou, Hang Liu, Jinbo Bi, Caiwen Ding, and Sanguthevar Rajasekaran. 2021. Against membership inference attack: Pruning is all you need. *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI'21)* (2021), 3141–3147.
- [26] Yijue Wang, Chenghong Wang, Zigeng Wang, Shanglin Zhou, Hang Liu, Jinbo Bi, Caiwen Ding, and Sanguthevar Rajasekaran. 2021. Against Membership Inference Attack: Pruning is All You Need. In *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI-21)*, 3141–3147.
- [27] Alexander Warnecke, Lukas Pirch, Christian Wressnegger, and Konrad Rieck. 2023. Machine unlearning of features and labels. In *Proceedings of the 30th Network and Distributed System Security Symposium (NDSS'23)*.
- [28] Jing Wu and Mehrtash Harandi. 2024. Scissorhands: Scrub Data Influence via Connection Sensitivity in Networks. In *Proceedings of the 18th European Conference on Computer Vision (ECCV'24)*, 367–384.
- [29] Yinjun Wu, Edgar Dobriban, and Susan Davidson. 2020. Deltagrad: Rapid re-training of machine learning models. In *Proceedings of the 37th International Conference on Machine Learning (ICML'20)*. PMLR, 10355–10366.